



Republic of the Philippines
CAVITE STATE UNIVERSITY
Don Severino delas Alas Campus
Indang, Cavite

COLLEGE OF ENGINEERING AND INFORMATION TECHNOLOGY

INTELLI-CANE UMBRELLA

Human Computer Interaction 2

Submitted to Aimiel Peña
College of Engineering and Information Technology
Cavite State University
Indang, Cavite

Laboratory Requirement
for the course Introduction to Computing
Bachelor of Science in Information Technology

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1. INTRODUCTION

Assistive technology helps improve the daily lives of people with disabilities, especially those who are visually impaired. It allows them to move and perform activities with more independence and safety. In the Philippines, sudden weather changes are common, especially rainfall, which can affect outdoor activities. Visually impaired individuals usually need to carry both a cane and an umbrella if it rains, which can be very difficult to use at the same time. This situation creates inconvenience and may affect their safety while traveling.

1.1 Product Brief/Overview

This project focuses on the development of the Intelli-Cane Umbrella, an innovative smart assistive device designed to support visually impaired individuals by combining the functions of a traditional cane and an umbrella into a single product.

The Intelli-Cane will be equipped with a hydro-sensor on top of the umbrella to sense when the rain has stopped, allowing the user to close the umbrella confidently. Additionally, a proximity sensor will be installed to help the user detect nearby objects or people, enhancing safety and navigation when the umbrella is close and will be used as a cane. The user will be alerted using a built-in speaker to help inform the user.

A smart band mechanism will also be included in the design. When the umbrella is closed, the smart band will automatically wrap around the umbrella to secure it in place, making it easier for the user to carry and manage the device.

Overall, the Intelli-Cane Umbrella aims to provide safety, convenience, and independence for visually impaired individuals.

1.2 Objectives

1.2.1 The main objective of this project is to develop the Intelli-cane Umbrella, a smart umbrella that is designed to assist visually impaired individuals during daily activities by combining the functions of an umbrella and cane into a single device.

The specific objective of this product is;

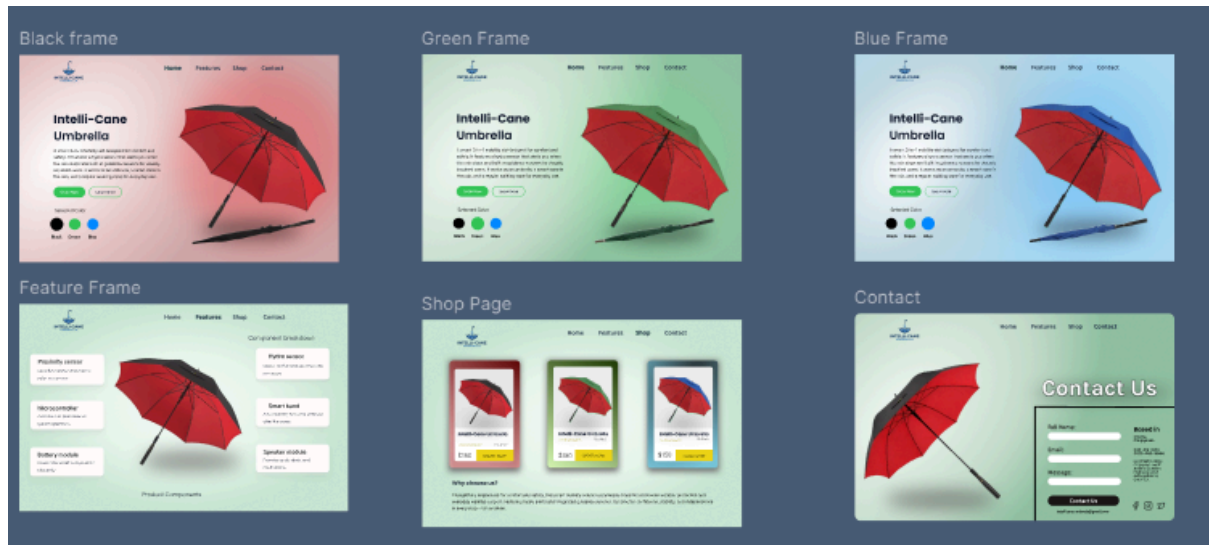
- To create an umbrella with a hydro sensor that can detect when the rain has stopped.
- To add an alarm or notification system that will alert the user when they can close the umbrella.
- To add a proximity sensor that can detect nearby obstacles when the umbrella is closed.
- To lessen the need for users to carry separate devices.
- To help users become more independent and confident to explore on their own.

2. TARGET USERS

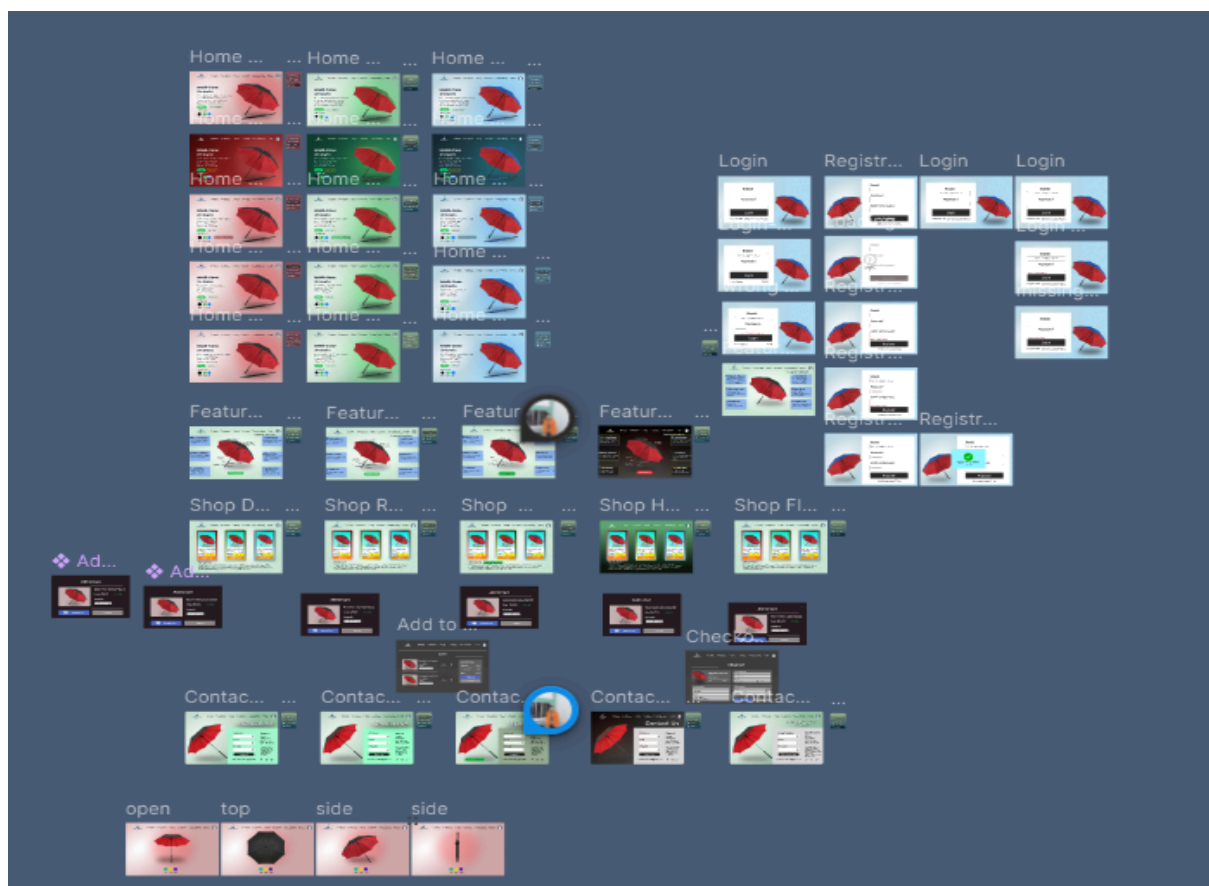
The primary target users of the Intelli-Cane umbrella are the visually impaired people because most of the visually impaired people use canes as assistance for performing daily activities; however, bringing a cane and an umbrella at the same time may be difficult for them, especially in the rainy season. Additionally, we think that they might want to experience functions like how normal people would, and sometimes they might want to explore on their own.

3. DESIGN AND PROTOTYPING

3.1. Initial Wireframe



3.2. Final UI Design



4. Technical Architecture

3.1. System Life Cycle

The Intelli-Cane Umbrella will work through a simple system process that helps the user in different situations.

First, the user will open the umbrella and turn on the system. After turning it on, the device will give a voice feedback such as “shade mode” to inform the user that the system is already running. The feedback will be delivered using a speaker.

When the system is on, the hydro-sensor and proximity sensor will start working. If it is raining, the hydro-sensor will detect it and inform the user through the speaker. It will also notify the user when the rain has already stopped so the user will know when it is safe to close the umbrella.

While the system is running, the proximity sensor is already active. It will detect if there are obstacles around the user and give a warning or heads-up through sound. This helps the user avoid accidents while walking.

When the umbrella is closed and the system is still on, the proximity sensor can still be used. In this mode, the device will act like a smart cane that can still detect obstacles in the user’s environment.

The system will also notify the user when the battery is low and needs to be charged using a Type-C port. The user can also adjust the volume of the speaker depending on their preference.

When the user closes the umbrella, the smart band will automatically wrap around the umbrella to secure it and make it easier to carry.

Overall, the system works continuously by giving information about rain, obstacles, and device status to help the user move safely and independently.

5. Timetable

Tasks	Specific explanation
Tasks Breakdown & Planning	The group first planned the overall design and concept of the Intelli-Cane Umbrella. The researchers discussed the features, appearance, and functions of the product before creating the low fidelity wireframe and Figma prototype. The materials and components needed for the physical prototype were also identified during this phase.
Timeline Structure	The development process was divided into phases which included planning, wireframing, Figma designing, prototype creation, testing, and final revision. Each task was scheduled properly to ensure the completion of both the digital design and physical prototype within the required timeframe.
Role Assignment / Team commitment	Each group member was assigned specific responsibilities to ensure equal participation during the development of the project. Some members focused on designing the Figma prototype, while others handled

	documentation, physical prototype preparation, research, and testing of the product concept.
Milestone & Deliverables	The group completed several expected outputs during the project development including the initial wireframe, final Figma design, physical prototype, revised design improvements, and complete project documentation. These outputs served as the basis for monitoring the progress of the project.
Resource Planning (Materials and tools)	The researchers used different materials and tools for the development of the Intelli-Cane Umbrella such as Figma for interface and product design, presentation materials for documentation, and prototype materials for the physical model. Internet resources and related studies were also used to support the project development process.

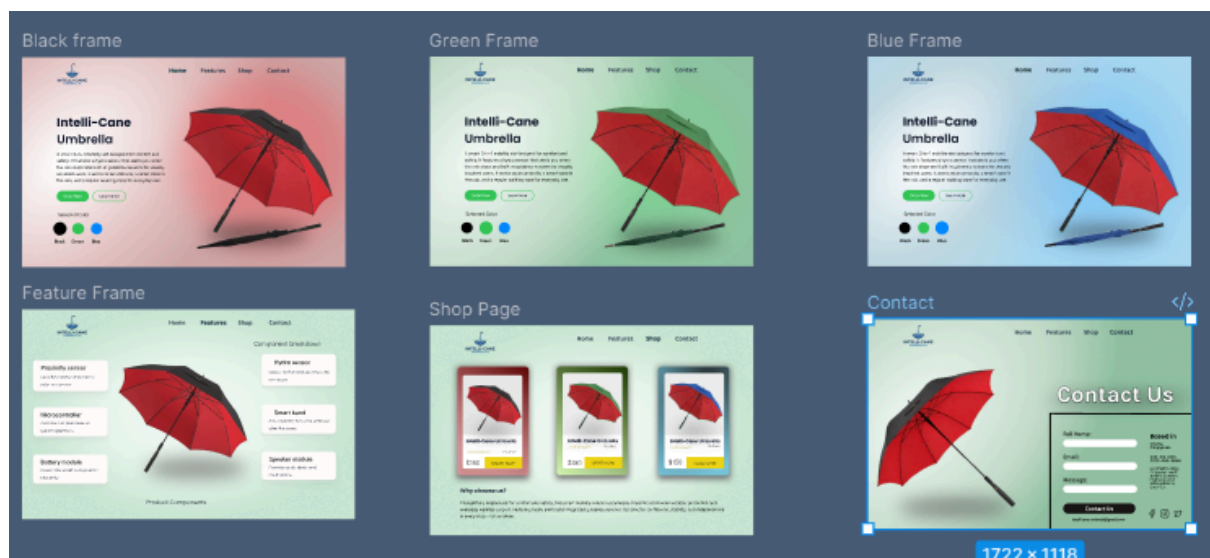
The timetable served as a guide for organizing the development process of the Intelli-Cane Umbrella. The researchers followed a step-by-step process beginning from planning and wireframing up to the creation of the Figma design and physical prototype. Proper task distribution, scheduling, and resource preparation helped the group complete the project efficiently while ensuring that the design and prototype met the intended objectives of the study.

6. Problems Encountered while developing the low fidelity wireframe and high fidelity wireframe

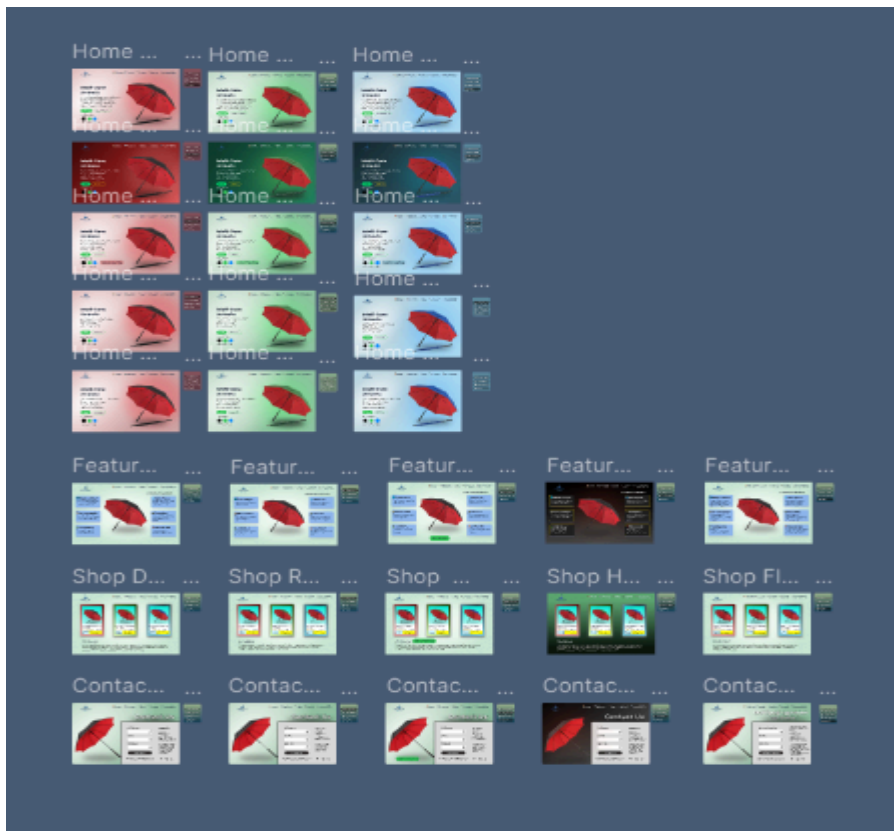
During the initial planning stage of the prototype, the researchers experienced difficulty in organizing the layout and identifying the proper placement of the features of the Intelli-Cane Umbrella. Since the product combines the functions of a cane and an umbrella, the group carefully discussed how the design would remain simple, accessible, and understandable for visually impaired users.

During the development of the prototype, the researchers experienced difficulties in arranging the layout, improving the design consistency, and organizing the features of the Intelli-Cane Umbrella in Figma. Several revisions and adjustments were made to improve the overall appearance and usability of the prototype. The group resolved these issues through collaboration, continuous editing, and design improvements.

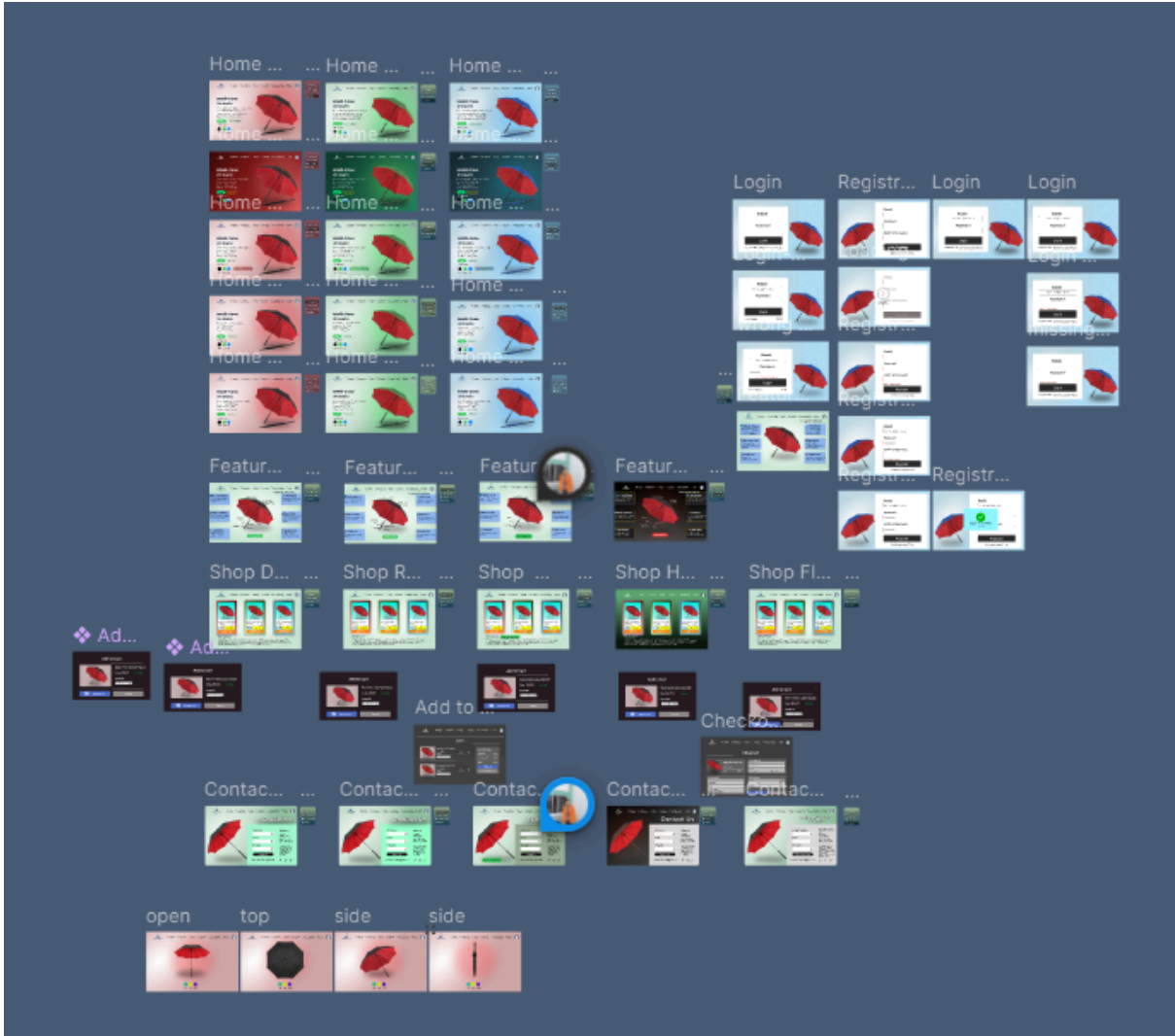
Initial Design (early layout / concept)



Revised Design (after adjustments / improvements)



Final Design (finished polished version)



LAB 7

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DSIT 3-1

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April 28, 2026

Saballo, Marc Christer I.

Product Brief:

- Intelli-cane Umbrella
- to help visually impaired people navigate when it's raining. the umbrella will give a heads up signal whether the rain stop pouring using sensor and speaker. Additionally, it can serve as a cane and can be use to detect people using sensor ~~tho~~ through their temperature under certain vicinity.
- Visually impaired people because we think they might want to experience function like how normal people would and sometimes they might want to explore on their own.
- ~~The~~ ~~product~~ HCI focuses on the interaction between the user and the system. It plays a significant role in our intelligence umbrella by indicating its importance to its user because it has functions that cater to what the user wants and needs. It highlights even more the benefits the user will gain because of how it is ~~made~~ designed.

Material List:

No.	Materials	Estimate Cost	Reason
1	Long Umbrella	₱ 300 - ₱ 400	It's suitable for modification based on our goal of how we want our product to be with additional functions. Because it is the one who detects when will the rain stop.
2	Hydrosensor	₱ 500 - ₱ 1000	Because it will be the one who will control all the functions of our product needed.
3	Microcontroller	₱ 1000 - ₱ 1500	Because it detects the people around the one who will use our product by a their temperature.
4	Proximity Sensor	₱ 1000 - ₱ 5000	Because it will make easier to our user to use our product by automatically wrapping the umbrella when they close it.
5	Smart Band	₱ 300 - ₱ 1000	Because it alerts the user to what was happening like when the rain already stop and when there are shade or people around.
6	Compact Speaker	₱ 500 - ₱ 700	Because it will supply the energy the intelligence umbrella needed to be able to make the functions work.
7	Battery	₱ 500 - ₱ 1000	Because it allows the product to be reused and make the function work again also type c it more convenient in present.
8	Type c charging Port	₱ 300 - ₱ 700	
		Total = ₱ 4400 = ₱ 11,300	

Product Estimate Cost = ₱ 5000

Prototype List

Prototype List	Estimate Cost
• Umbrella	₹ 200
• Microcontroller	₹ 100
• Illustration Board	₹ 100
• Magnet	₹ 50
• Paper	₹ 100
• Glue stick	₹ 50
	₹ 500

Build timeline

Activity	Week 1	Week 2	Week 3
Modification and Adding features			
Testing and Improving			
Polishing and finalized			

User Interaction Plan

- Open the umbrella ^{intelligently} to turn on the system.
- After the turning on the system, it will give a text status feedback like shade mode.
- If the system detected that it is raining, it will give a notification to the user and also when the rain already stop.
- Additionally, while the system is running, the proximity sensor is already at play, it will detect whether there is an obstacle around the user environment and will give a heads up.
- Proximity sensor can also be used when the umbrella is closed as long as the system is on making it looking like a smart cane.
- When the system runs out of energy, you can change usage. It will notify the user to change it using a type C port.
- You can also manipulate the speaker if you want to increase or decrease the volume.
- Smart band works when the you close the umbrella as it will automatically wrap around the umbrella.

Sketch & Blueprint

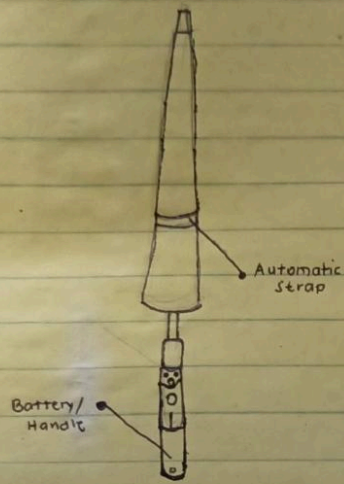


FIGURE 1: FRONT VIEW / CLOSED

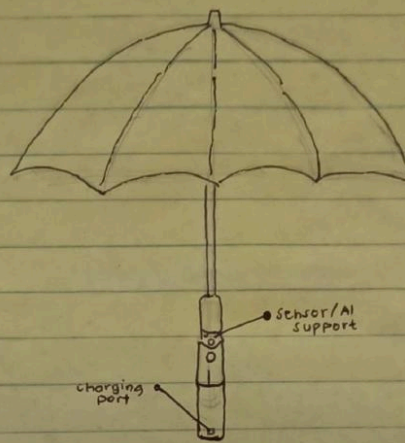


FIGURE 2: FRONT VIEW / OPEN

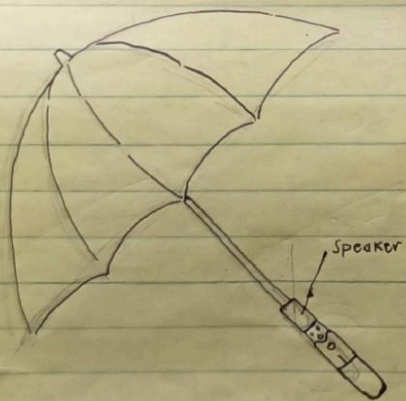


FIGURE 3: SIDE VIEW

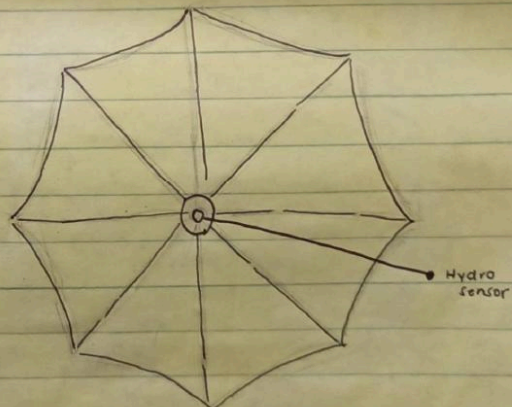


FIGURE 4: TOP VIEW

Laboratory RUBRICS

Criteria	Exceptional (100%)	Proficient (80%)	Developing (60%)	Needs Improvement (40%)
Introduction & Project Overview (10 pts)	Provides a comprehensive and well-contextualized explanation of the product; clearly explains purpose, scope, and relevance with strong technical grounding.	Clear explanation of the product and its purpose; lacks deeper context or technical justification.	Basic overview is present but lacks clarity or depth; minimal explanation of purpose.	Only states the product title or gives a vague, unclear description.
Objectives (10 pts)	Objectives are clearly explained, specific, measurable, and aligned with the system; well-justified with strong linkage to the project goals.	Objectives are clear and relevant but lack full detail or measurable aspects.	Objectives are present but vague or loosely connected to the project.	Objectives are missing, unclear, or not aligned with the system.
Target Users (10 pts)	Thorough identification of users; includes detailed justification, user needs, and relevance to the system design.	Identifies target users and provides general reasoning.	Mentions users but lacks explanation or justification.	No clear identification of target users.
Initial Wireframe (10 pts)	Complete and well-organized wireframes; clearly demonstrates layout planning and user flow.	Wireframes are present and understandable but lack detail.	Wireframes are incomplete or poorly organized.	No wireframe or unclear visual representation.
Final UI Design (10 pts)	High-quality UI screenshots showing a polished, user-friendly design; strong consistency and usability principles applied.	UI design is complete and functional; minor issues in consistency or clarity.	UI is present but lacks polish, consistency, or usability considerations.	Missing or very poor UI design presentation.

System Life Cycle / Process Flow (15 pts)	Detailed and accurate explanation of system processes/modules ; clearly explains how the system works step-by-step.	Explains system flow clearly but lacks deeper technical detail.	Basic explanation is present but skips key steps or lacks clarity.	Incorrect or missing explanation of system flow.
Technical Architecture (15 pts)	Provides a well-structured and detailed explanation of system architecture; clearly explains modules, components, and interactions.	Architecture is explained clearly but lacks depth or technical detail.	Basic structure is described but unclear or incomplete.	No clear explanation of system architecture.
Timetable (10pts)	Provides a well-defined, detailed, and realistic timetable showing clear execution of developing the physical prototype; includes task breakdown, timeline, roles, milestones, and resource planning with clear explanations.	Timetable is well-defined; includes most required components but has minor gaps in detail, explanation, or realism.	Timetable is present but lacks detail and clarity; some components are missing or are poorly explained.	Timetable is incomplete, unclear, or missing; lacks proper planning, structure, and explanation of how the prototype will be developed.
Problems Encountered & Documentation (10 pts)	Clearly explains multiple challenges with detailed solutions and supporting screenshots; demonstrates strong problem-solving.	Explains problems with some supporting evidence; solutions are clear.	Mentions problems but lacks detail or supporting proof.	No meaningful discussion of problems encountered.